

The empirical recurrence for OEIS A231936, recovered from its tail

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Abstract

OEIS A231936 records “Empirical recurrence of order 65” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 98$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 67$. Its last coefficient is $c_{65} = -25743592348976362291200 \neq 0$, so no further tail satisfies anything shorter; any order-65 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A231936 is “Number of nX3 0..3 arrays with no element less than a strict majority of its horizontal, diagonal and antidiagonal neighbors.”. It has offset 1 and begins

16, 668, 8426, 152180, 2462572, 36885538, 586971925, 9394148948, ...

The entry states:

Empirical recurrence of order 65 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:44:09, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 98$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 98$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 65: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 65.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 196$ exact terms from $n = 2$ on, it returns order 65 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{65} c_i a(n-i),$$

c_1	16	c_{18}	216719403239162	c_{35}	239714859137879351840	c_{52}	120336184550378
c_2	−72	c_{19}	−198158054384185	c_{36}	17796585998020360896	c_{53}	62867712682786
c_3	2700	c_{20}	−396696596946587	c_{37}	−729279133875667938888	c_{54}	12213109219164
c_4	−22020	c_{21}	−10500971430765614	c_{38}	−3155023580485704357008	c_{55}	−38490418181992
c_5	65604	c_{22}	−4711029840464083	c_{39}	−1699064648307790563344	c_{56}	−1128627501086
c_6	−2360209	c_{23}	−24950476545214369	c_{40}	−4443447088815112199008	c_{57}	−16047781363536
c_7	7196789	c_{24}	134712481276481106	c_{41}	5043348516714926709056	c_{58}	−7406764335546
c_8	−43952301	c_{25}	339098876521525925	c_{42}	12620127354734150816896	c_{59}	24502188304744
c_9	749959726	c_{26}	690226440628323919	c_{43}	12675735972315319320384	c_{60}	−5803108755456
c_{10}	−416181928	c_{27}	803710983666293904	c_{44}	37306573931453775288576	c_{61}	111884365434008
c_{11}	12897064247	c_{28}	−3448875715840945162	c_{45}	−24209922517944890509056	c_{62}	17821643609454
c_{12}	−82692710461	c_{29}	−7468899913816504523	c_{46}	−5012287243312531290624	c_{63}	−5309518177739
c_{13}	−84006603600	c_{30}	−22273545875537860333	c_{47}	−84693846831756126686208	c_{64}	20853981464511
c_{14}	−1611383942755	c_{31}	−16550621796169785089	c_{48}	−108664779745869184871424	c_{65}	−2574359234897
c_{15}	1081180492576	c_{32}	32256886470403308352	c_{49}	29177927674436302331904		
c_{16}	7129671818649	c_{33}	101166036874738922912	c_{50}	−64710635606038433894400		
c_{17}	64510215193979	c_{34}	342930238659215051576	c_{51}	265846188415766798057472		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 67$, and it is the recurrence OEIS A231936 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{65} - c_1 x^{64} - \dots - c_{65}$. Its constant term is $-c_{65} = 25743592348976362291200$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no

component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 65 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 65, so $\deg q = 65$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 65, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 65, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -25743592348976362291200 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-65 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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